

Laurea Magistrale in Ingegneria Informatica
a.a. 2024-2025

Scalable AI

Introduction

Lorenzo Baraldi, Giuseppe Fiameni

Welcome!

- This is the course of “**Scalable AI**”.
- **Course TLDR**; *Everything you wanted to know about Large-Scale Foundation models, High Performance Computing (HPC), and Deep Learning at scale. And more.*
- Typically, we would have classes **each Monday** and **each Thursday**, even though there might be variations.
- Keep yourself updated and **sign up on the course website (Moodle)** to get e-mail notifications.

Your instructors

- **Prof. Lorenzo Baraldi**

Dipartimento di Ingegneria «Enzo Ferrari»

e-mail: lorenzo.baraldi@unimore.it

Phone: 0592056353

Office hours can be booked at <http://ailb-web.ing.unimore.it/lorenzo/ricevimento>

- **Dott. Giuseppe Fiameni**

NVIDIA

e-mail: gfiameni@unimore.it

Office hours: book via e-mail

Hardware and architectural foundations for large-scale AI

- CPU vs GPU processing flows
- Multi-GPU architectures
- Multi-Node architectures
- Going parallel: data, model, pipeline, tensor parallelism

State-of-the-art Deep Learning architectures for Computer Vision, Natural Language Processing and Multimedia

- The Attention operator, self/cross-attention and its variants
- The Transformer architecture
- Transformer-based models for Language (BERT, BART, GPT-x, LLaMA2)
- Transformer-based models for Computer Vision (ViT, DeiT, DETR)
- Usage of learnable queries, Perceiver, Perceiver.io
- Connecting multiple modalities: encoder-decoder, encoder-only and decoder-only approaches
- Optimised variants of the attention operator

Syllabus (2/3)

Self- and semi- supervised training approaches for large-scale data

- Masked language models, Next Sentence prediction
- Noisy labels / Self-training
- Contrastive learning
- Multi-modal foundation models: CLIP, Flamingo
- Scaling laws for Deep Learning architectures
- The Mixture of Experts approach

Distributed training in the PyTorch environment

- PyTorch data parallel
- Distributed Data parallel
- The torchrun utility
- Advanced DDP libraries: horovod, deepspeed
- Libraries and SOTA solutions for managing large-scale datasets

Syllabus (3/3)

Scalable techniques for large-scale models

- SGD for large batches, Asynchronous SGD
- Mixed Precision training
- Zero Redundancy Optimizer(s)
- Managing training with large batches
- QAT, PTQ

Introduction to HPC environments

- Architecture of a supercomputer
- High-speed network protocols, distributed file systems
- Managing resources and the queue: the SLURM queue manager
- Practical cases: a small scale (lab-size) cluster, and a supercomputer, e.g. LEONARDO
- Interactive sessions (e.g. process affinity, task affinity)

Teaching modality

- The course is offered in presence (no, we do not stream)
- We want lectures to be as interactive as possible
 - You can/should ask questions, do it! 😊
- Slides will be uploaded regularly on Moodle
- Handouts and textbooks should be considered as a starting point and a point of reference, but they cannot be the limit beyond which one cannot go. The phrases "this is not in the text" or "he did not explain this in class" are not an excuse during the exam.

Getting support

How do I avoid getting stuck? What should I do if I actually get stuck?

Ask for support

- Do not hesitate to reach out and ask for support.
- In case it is needed, we can also give you a tutor (either a PhD student or a research grant) who can give some tutorship.
- Still: always recall that you are many.
 - Try to be autonomous and independent.
 - *Do not* ask support for simple issues that you can solve by your own.
 - *Do not* ask support on something that has already been covered in previous courses.
 - Contact the teacher/tutor when it is really needed, or when you need advice from an expert.

Taking the exam

- **The exam is oral.** You will be asked different questions, covering all topics seen during the lectures.
- There is no practical project.
- We will regularly add and advertise exam sessions on ESSE3.
- *Enrolment on ESSE3 closes at the beginning of the day before the exam; i.e., if an exam is scheduled for June, 2nd at 9:00, registration will close on June 1st 00:00 (24+9 hours before the exam session).*
- If you don't book your exam on ESSE3, you will not be allowed to take the exam.
- After the exam, you will receive an e-mail with your grade. You can accept or refuse it, and take the exam again.

Finally, if you have any doubts, do not hesitate to reach out and ask for clarifications.